Alexander **Zhuk**

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Education

Illinois Institute of Technology *Chicago, IL*

B.S. in Civil Engineering GPA 4.0 *Expected May 2029*

Skills

CAD	SolidWorks | AutoCAD | Inventor | Revit | Blender
Software	Ansys Fluent | Visual Studio Code | MATLAB | Unity
Fabrication	Lathe Machining | Mill Machining | Soldering | Metal working
Digital Media	Photoshop | Lightroom | Davinci Resolve | Microsoft 365
Languages	English | Ukrainian | Russian

Experience

IIT Motorsports (Formula SAE) *Chicago, IL*

Vice President & High Voltage Team Lead *August 2025 to Present*

- Directed workflow and project timelines for a 50-member team Formula SAE team, establishing a comprehensive 2-year vehicle build schedule.
- Led a 6-person specialized sub-team in the end-to-end design and mechanical integration of a 2026 electric tractive battery system.

AZcore (FSAE) *Chicago, IL*

LARP (Lithium Array Rechargeable Pack) *August 2026 to Present*

- Engineered a modular multi-plate battery assembly utilizing standardized commercial off-the-shelf components, streamlining machining and assembly cycles while increasing mechanical performance compared to previous iterations.
- Optimized battery module architecture using SolidWorks, successfully increasing pack density by 6% and overall performance by 4%.
- Developed and validated a custom active cooling system for a 72s6p accumulator module using ANSYS Fluent Conjugate Heat Transfer (CHT) simulations, reducing operating temperatures by 5°C and boosting thermal efficiency by 7%.

American Society of Civil Engineers Steel Bridge *Chicago, IL*

Fabrication Team Member *August 2025 to Present*

- Manufactured structural steel components for the competition bridge, utilizing shop tools and mill machinery to meet precise design specifications.
- Collaborated with 6 team leads to review bridge design to ensure all components were properly drafted.

Structural Engineers Association of Illinois *Chicago, IL*

VP of Membership *December 2025 to Present*

- Drafted a comprehensive 2026-2027 events calendar, including technical workshops, site visits, and structural design competitions to increase member engagement.

Plainfield North Theater *Plainfield, IL*

Lead Structural Designer & Sound Tech Lead *August 2023 to May 2025*

- Directed structural design and construction of multi-floored theatrical sets to safely support a 30+ member cast
- Managed live signal levels and channel activation for 20+ wireless microphones, ensuring precise audio cues and consistent gain staging throughout multi-act performances

Involvement

Project Teams	IIT Motorsports (FSAE) | Steel Bridge
Prof. Societies	SEAOI | ASCE | SHPE
Campus Org	Delta Tau Delta (ΔΤΔ) | Hawk Ambassador | Exelon Summer Institute
Media	Photography Club